Sambhav Surana

AI/ML Developer | Python Developer

I build AI agents, computer vision systems, and full-stack ML products that turn research ideas into working software.



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Experience

Abhyudaya Coding Club
Club Member
Shri Vaishnav Vidyapeeth Vishwavidyalaya
2024 - Present

- The development of AI-based projects including Time Table Generation system using constraint satisfaction algorithms
- Organized workshops on Python, NumPy, Pandas, and Scikit-learn for club members
- Collaborated with team members on model optimization and performance enhancement

Certifications

Hackathons & Competitions
Hackwave 2.0
Aug 2025

Participated in a 36-hour coding hackathon organized by Echelon Dev Society at CDGI, Indore. Gained hands-on experience in rapid prototyping, teamwork, and problem-solving.

https://hackwave-site.vercel.app/

Introduction to Machine Learning
NPTEL Online Certification - IIT Kharagpur
Jul-Sep 2024

- Achieved Elite Certificate with a consolidated score of 66%
- Completed an 8-week course covering fundamental machine learning concepts and assignments

Technical Skills

Programming Languages
Python, JavaScript, SQL, Bash

ML Libraries & Frameworks
NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib

Computer Vision
OpenCV, Image Processing, Data Augmentation, YOLO, Face Recognition, Object Detection

Deep Learning
CNN, Neural Networks, Model Training, Transfer Learning, Feature Extraction

NLP & Generative AI
LangChain, OpenAI API, Ollama, Text Processing, Query Decomposition, LLM Integration

Deployment & Tools
REST APIs, Git, GitHub, Model Deployment, Gradio, WebSockets

Databases
MySQL, MongoDB, ChromaDB

Web Technologies
React.js, Next.js, HTML/CSS, Tailwind CSS

Development Tools
VS Code, PyCharm, Jupyter Notebook, Git

Operating Systems
Windows, Linux

Education

Shri Vaishnav Vidyapee...
CSE (AI-IBM) III Year
7.02 CGPA
B.Tech.
2023 - Present

Projects

Activation-Weighted Low-Rank Compression

Research on activation-weighted low-rank compression of transformer weights, achieving 5-7x improvement over plain SVD.

2026

- **Method:** Closed-form activation-weighted fitting that minimizes error in activation space rather than weight space.
- **Results:** Reduced perplexity degradation from +21% to +3.6% (GPT-2), +72% to -4.5% (Gemma-3), +26% to +3.8% (Qwen2.5).
- **Storage:** int8 factor quantization proven lossless at 3x per-matrix compression ratio.

Python, PyTorch, Transformers, Research

https://github.com/Sambhav242005/Weighted-Lowrank-Compression

Velora Transformer Training

A local-first LLaMA-style training stack for safe small-model pretraining and SFT workflows.

2026

- **Training Infrastructure:** Built RoPE, RMSNorm, SwiGLU, GQA, checkpoint safety, and restartable training loops.
- **Data Pipeline:** Prepared tokenizer, FineWeb-Edu, and SFT workflows with RAM-safe memmap loading.
- **Run Safety:** Added micro-batch search, VRAM logging, and atomic checkpoint writes for interrupted runs.

Python, PyTorch, Transformers, FineWeb

https://github.com/Sambhav242005/Velora

Reelbox

Local AI library for saved Instagram reels with topic organization and semantic search.

2026

- Built a Chromium extension and Python dashboard to collect saved reel links and browse topic folders.
- Integrated local multimodal analysis with MiniCPM-V through Ollama and embedding-based semantic retrieval.
- Persisted analysis queues, stop/resume state, verification pauses, and manual folder assignments across restarts.

Python, Ollama, MiniCPM-V, Chromium Extension

https://github.com/Sambhav242005/Reelbox

AI Face Detect and Recognition WebGPU

A client-side AI face tracking and recognition application running YOLOv11 and EdgeFace directly in the browser via WebGPU and ONNX Runtime.

2025

- **Browser-Based AI:** Ran YOLOv11 face detection and EdgeFace embeddings in-browser via WebGPU/WASM ONNX Runtime.
- **FastAPI Backend:** Designed a lightweight backend to host FAISS session indexes and JSON metadata.
- **Features:** Built session persistence, pause/resume tracking states, and real-time cropping pipelines.

WebGPU, ONNX Runtime, YOLOv11, FastAPI, FAISS

https://github.com/Sambhav242005/AI-Face-Detect-and-Recognition

Languages

English

Hindi

German